

Tesla Stock Price Analysis Report

Tesla Inc. (NASDAQ: TSLA) | 2024–2026



UNIVERSITY OF COLOMBO

Faculty of Science

Industrial Statistics and Mathematical Finance

Submitted by

A.H. Thisara Bhagya Anupama

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1. Introduction

For this project, I selected the stock of Tesla Inc. (Ticker Symbol: TSLA). Tesla is one of the most recognized electric vehicle and technology companies in the world. The company is known for innovation in electric vehicles, renewable energy, and battery technology. I selected Tesla because its stock prices show significant movement and volatility, making it suitable for financial analysis and risk measurement. Tesla is also widely followed by investors and financial analysts, which makes it an interesting company for stock market analysis. The purpose of this report is to analyze Tesla's stock performance using historical stock market data and evaluate the return and risk associated with the investment.

2. Data Description

The stock market data was collected from Yahoo Finance using the Python yfinance library. The dataset includes daily stock price information such as opening price, closing price, highest price, lowest price, and trading volume.

Dataset Summary:

- Company: Tesla Inc.
- Ticker Symbol: TSLA
- Data Source: Yahoo Finance
- Date Range: 2024-01-02 to 2025-12-30
- Number of Trading Days: 501
- Variables Used: Open, High, Low, Close, Volume No missing values were found in the dataset, which indicates the data was complete and suitable for analysis.

3. Charts and Analysis

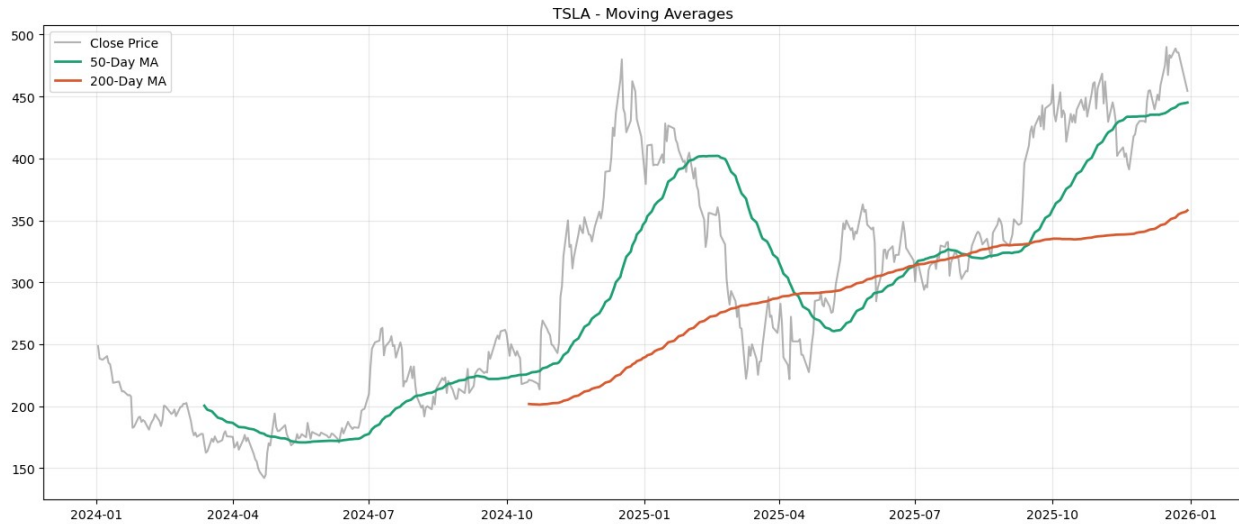
Closing Price Trend:

The closing price chart showed significant fluctuations during the selected period. Tesla stock experienced both upward and downward trends, reflecting the volatile nature of technology stocks.



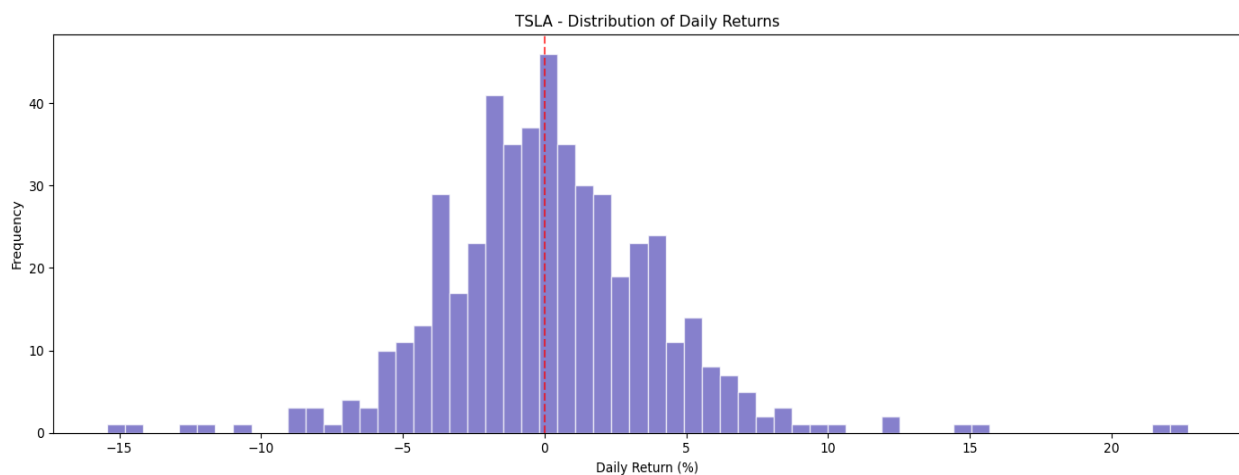
Moving Average Analysis:

The 50-day and 200-day moving averages helped identify long-term and short-term trends. When the 50-day moving average stayed above the 200-day moving average, the market showed bullish behavior.



Distribution of Daily Returns:

The histogram of daily returns showed that most returns were concentrated near zero, while several extreme positive and negative returns were observed. This indicates high volatility and investment risk.



Cumulative Return Analysis:

The cumulative return chart showed overall positive long-term returns despite temporary declines. This suggests Tesla provided profitable growth opportunities during the study period.



4. Risk Metrics

Sharpe Ratio = 0.71

The Sharpe Ratio measures risk-adjusted return. A value of 0.71 indicates moderate returns relative to the level of risk taken.

Maximum Drawdown = -53.77%

This measures the largest decline from a previous peak. Tesla experienced a major temporary decline, indicating substantial downside risk.

Annualized Volatility = 63.56%

The high volatility value shows that Tesla stock prices fluctuated significantly during the selected period. This reflects high investment risk and uncertainty.

Annualized Return = 50.34%

It measures how much profit or loss an investment generates over one year on average. It helps investors understand the overall yearly performance of a stock and compare it with other investments.

Tesla had an annualized return of 50.34% , which means the stock generated a strong average yearly return during the selected period.

5. Conclusion

This project analyzed Tesla Inc. stock performance using historical stock price data from Yahoo Finance. The analysis showed that Tesla generated strong overall returns during the selected period. The stock demonstrated high growth potential and positive cumulative returns. However, the stock also experienced large price fluctuations and significant drawdowns. Overall, Tesla can be considered a high-risk, high-return investment. Investors who are willing to accept higher risk may benefit from potential long-term returns, while conservative investors may find the volatility too high. In conclusion, Tesla was a profitable investment during the selected period, but it involved considerable market risk.
